

SELENIUM

Day 1

Introduction to Java:

→ Java is a programming language used for

1. web applications
2. Automation Testing
3. Android Applications
4. Enterprise Software

→ Java features:

1. Platform Independent
2. Object oriented
3. Secure
4. Robust.

Requirements:

Java 17 SDK

chrome

chromeDriver } same version

IntelliJ IDEA ✓

(or)
Eclipse

→ JDK vs JRE vs JVM

1. JDK : used to develop java applications
2. JRE : used to run java applications
3. JVM : converts bytecode into machine code

→ Java program flow

Java code (.java)



compiler (javac)



Bytecode (.class)



JVM



Machine code.

→ Variables and Data Types

- variables are used to store the data.

Testing:

→ Slw Testing checks whether an application works correctly.

→ Types of Testing:

1. manual Testing
2. Automation Testing.

Selenium:

→ Selenium is a tool used to automate web applications

→ components of selenium:

1. Selenium IDE
2. Selenium WebDriver
3. Selenium Grid.

Selenium Architecture

Java code



Selenium WebDriver



ChromeDriver



Chrome Browser.

What is Maven?

→ Maven is a build automation and dependency management tool.

→ It is like a playstore. In real life we used to download applications like whatsapp from the playstore. and playstore manages all the backend activities and dependencies and helps to download the whatsapp.

In the same scenario maven used to manage all the dependencies and activities regarding S/W Testing & projects.

→ here pom.xml is the main Maven configuration file.

It is used to add dependencies
configure projects
manage builds

Selenium Dependency:

<dependencies>

<dependency>

<groupId> org.seleniumhq.selenium </groupId>

<artifactId> selenium-java </artifactId>

<version> 4.21.0 </version>

</dependency>

</dependencies>

First Selenium program:

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class FirstTest
{
    public static void main(String[] args)
    {
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");
    }
}
```

O/p:

It opens Google web.

→ web opens and close after some time.

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class FirstTest
{
    public static void main(String[] args) throws
        InterruptedException
    {
        WebDriver driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.get("https://www.amazon.in");
        System.out.println(driver.getTitle());
        Thread.sleep(3000); // closes after 3 seconds
        driver.quit();
    }
}
```

O/p: It opens amazon website in full window size and closes after 3 seconds.

Selenium methods:

- driver.get() → opens website.
- driver.getTitle() → Returns page title
- driver.getCurrentUrl() → Returns current URL
- maximize() → Maximized browser
- driver.quit() → closes browser
- { close() → closes current tab
- { quit() → closes entire browser